

UNIT V : DSP PROCESSORS :-

A DSP Processor is a specialised uP, targeted at DSP applications. It is a specialised chipset used in optimal processing of digital signal samples, performing repeated operations on each sample.

Salient Features :-

- (i) Multiplier & Multiplier Accumulator (MAC)
- (ii) Modified Bus Structure and Memory Access Schemes
- (iii) Multiple Access Memory
- (iv) Multiported Memory
- (v) Very Long Instr. Word [VLIW] Architecture
- (vi) Pipelining
- (vii) Special Addressing Modes
- (viii) On-chip Peripherals
- (ix) Harvard Architecture

Components :-

- ALU Arithmetic Logic Unit
- Multiplier (*)
- FPU Floating Point Processing Unit
- Registers Memory cells w/ data
- Accumulator Special Reg., holds immediate results
- Address Generator
- Buses Lines for data exchange

Different Processor Architectures :

Von Neumann

Harvard

(1) Same memory for program & data

Separate memory for program & data

(2) Common bus for all data xfer.

Separate buses for data & instrⁿ.

(3) '2' CLK Cycles for execⁿ of 1 instrⁿ.

Only 1 CLK Cycle required.

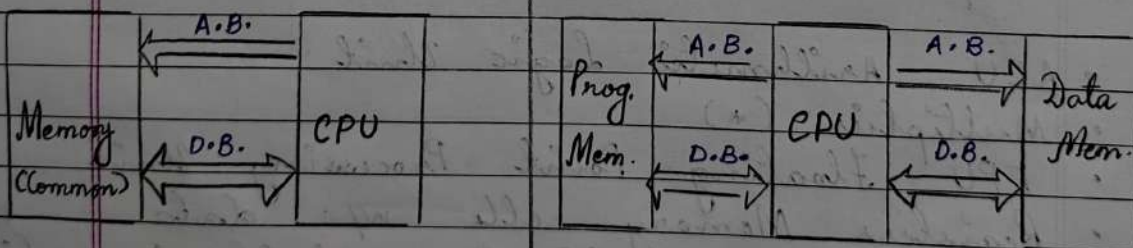
(4) Cheaper

Costlier

(5) Used in CPUs, of PCs

Used in DSP Processors

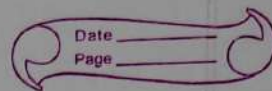
(6)



* Harvard Archi. extensions in DSP :

- Dual Memory Access
- Instrⁿ Cache
- I/p - o/p Controller

* Block Diagram of DSP [PDF, Pg. 15]



Linear Buffering :-

- 'N' latest samples are processed
- Samples are stored in a buffer.
- A new sample arrives:
 - Oldest sample removed
 - Remaining samples shifted by one
 - New sample placed at end.
- Uses FIFO Mechanism,
- Processor Cycles are wasted for reallocⁿ of samples.

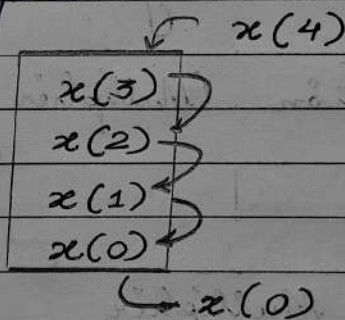
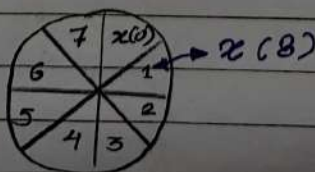


Fig. Queue of Instrⁿ.

→ Key Features of DS Processors :-

(1) Circular Buffering

- Pointer indicates current write position
- New sample written at x_{ter}
- x_{ter} advances to next position
- No data moved $\therefore$ Faster



- Efficient, only 1 operⁿ to perform a task.

* Pic. rep. of linear & circular addressing

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Circular Addressing

Practically, circular buffer is a linear buffer and a *ter.

- *ter is moved, indicates order of sample processing.
- Once *ter reaches end, wraps to start
- Linear buffer in memory uses circular addressing.

(2) MAC Operation (Multiplier Accumulator)

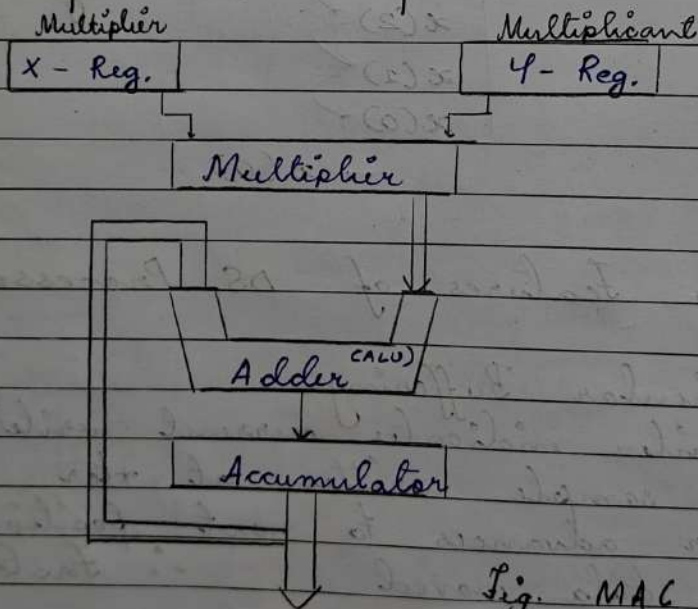


Fig. MAC Oper?

- A common DSP operation (eg. DFT, FFT etc.) requires multiplicⁿ & accumulation of result

$$y \leftarrow y + a * x$$

On a standard CPU, multiplicⁿ & additⁿ are done separately.

- ★ Therefore, DSP Processors have MAC implemented in H/W as a single processor instrⁿ.

- This reduces no. of used cycles and speeds up program execⁿ.

- ★ Many modern DSPs can do '2' MAC operⁿ at same time (dual MAC)

Advantages: • Determines power consumption and speed of overall system.

• High speed, easy computatⁿ.

(3) Pipelining

Let '4' stages of instrⁿ be: Fetch, Decode, Read/Write, Execute

Pipelining is performed as follows:

- First stage (F) of 1st Instrⁿ is performed.

• When next stage (D) is executed, first stage (F) of second instrⁿ is performed, and so on.

• Instrⁿ stages are performed with overlapping, speeding up execⁿ.

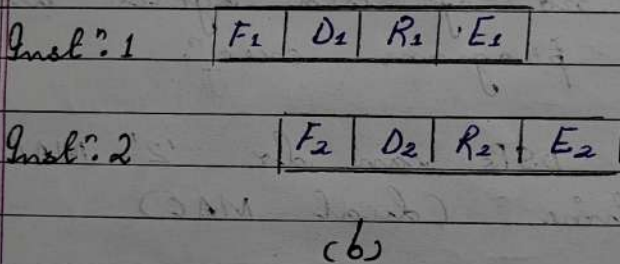
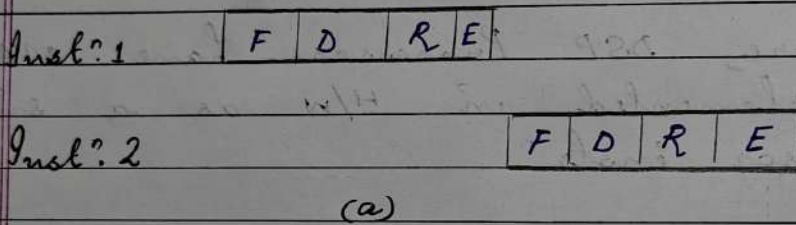


Fig. (a) w/o Pipelining
 C → 2 instrⁿ : 8 CLK

(b) w/ Pipelining
 C → 2 instrⁿ : 5 CLK

(4) Very Long Instrⁿ. Word (VLIW)
 It is a processor archi. that achieves high performance by having the compiler, rather than H/W, pack multiple independent operⁿ into a single, long instrⁿ for simultaneous execⁿ.

Adv.: • ↓ H/W Complex • ↓ Scheduling needs
 • ↓ Power Consum.

→ Fixed - Point v/s Floating - Point DSP Processor:

auto scaling by exponential rep

Feature	Fixed	Floating
Number Representat ⁿ	Integers w/ decimal point.	IEEE-754 style mantissa + exponent
Precision	Limited	High dynamic range
Range	Small	Very Large
Speed	Faster	Slower
H/W Comp.	Simpler ALU Design	More complex FPU Design
Power Con.	Low	High
Programming Difficulty	Harder, needs manual scaling	Easier, automatic scaling
Accuracy	Lower	Higher
Over/Underflow	Common	Rare
Eg.	TI TM320C55x, ADSP-218x	SHARC DSP

Central Processing Unit v/s

Graphics Processing Unit v/s

Neural Process. Unit

CPU v/s GPU v/s NPU

Feature	CPU	GPU	NPU
Purpose	General computing & system control	Parallel Processing, Graphics Rendering, Heavy Comp.	AI/ML Oper?
Architecture	Few powerful cores	Many smaller cores	Specialised AI cores (MAC, Tensor)
Processing	Sequential + Limited Parallel	Parallel	Parallel, optimised for neural ~10
Speed	Slowest	Faster	Fastest
Efficiency (Power)	Moderate	High	Very efficient for AI workload
Flexibility	Very flexible	Yes, for parallel tasks	Less
Typical Applic ⁿ	Running prog, browser, OS	Gaming, Video Editing	Voice Assis, AI Camera
Eg	Intel i7, i9, AMD Ryzen 7/9, ARM Cortex-A78	NVIDIA RTX 3060 / 4090, Intel Arc A770	Apple Neural Engine, Qualcomm Hexagon NPU

⇒ Super Harvard Architecture (SHARC)

SHARC is a high-performance DSP design from Analog devices that extends traditional Harvard Archi.

It improves throughput by combining separate instrⁿ. & data memories w/ an instrⁿ. cache and a dedicated I/O Controller

• Key Features :-

- ① Enhanced Harvard: Adds high speed instrⁿ. cache, preventing bottlenecks while accessing external memory
- ② I/O Controller: Integrates DMA controllers, allowing I/O operⁿ. to run indep. of CPU core.
- ③ Optimized Core: Optimized for MAC & zero-overhead looping.

IMP: (not mentioned)

• DS Processor Block DIAG.